

Puneet Ludu

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🌐 puneet.io | 📄 www.kaggle.com/puneetsl | 📄 scholar.google.com/citations?user=NrYKcaMAAAAJ&hl=en | 📍 New York, NY

Education

State University of New York

Master of Science in Computer Science

2014

JIIT

B. Tech. in Computer Science and Engineering

2010

[Link ↗](#)

Experience

Zillow (Zestimate)

Sep 2021 – Present

Machine Learning Engineer (Tech Lead)

Remote

- Led proof-of-concept and core architecture for Listing IQ CMA: an AI-powered tool enabling agents to customize property comparisons with map-based filtering, editable valuations, and amenity-level explanations. Designed APIs integrating real-time Zestimate valuations, property embeddings, and comparative analysis. Mentored junior engineer through production launch.
- Designed and shipped the first LLM assistant deployed to Zillow customer support: personalized Zestimate explanations grounded in real valuation data. Built RAG pipeline that constrains the agent to answer from actual Zestimate features and values rather than free-form generation; layered output validation with schema and range checks to catch out-of-distribution responses before they reach customers. Deployed with deflection rate tracked as the primary production KPI.
- Architected and built a similar listings comparison engine from the ground up: batch pipeline and real-time API using H3 geospatial indexing and customizable ranking algorithms. Processes 3M+ listings, powers Zillow Showcase dashboards (8M monthly views) helping agents demonstrate listing performance vs non-Showcase homes. Led migration to append-only listing data source; built lifecycle management and deduplication layer to track active, sold, and removed listings across markets. Production in 5 weeks. Later led algorithm improvements.
- Built Valuation API from POC to production: FastAPI service handling ~6K requests/day with zero P2 alerts since launch. Led Zestimate 6.6 deprecation (Redis→hybrid cache feature store for real-time valuations, legacy system shutdown), saving \$350K annually. Drove several Zestimate model point releases to completion; created MR tracking system, led ETL optimizations. Completed CI/CD pipeline modernization across all team services with zero production incidents.
- Leading Image Embeddings project: experimenting with DualLossAutoEncoder to predict property condition from CLIP embeddings (3.3M listings) for Zestimate integration. Prototyping a NAM-based (Neural Additive Models) Zestimate variant for intrinsic interpretability, alongside a per-feature attribution architecture that exposes dollar-level contributions to downstream AI agents. Led core Zestimate neural network ETL pipeline migration to Databricks; identified and resolved 8x latency and 10x cost regression before production deployment.
- Managed summer intern (2023): designed project plan, weekly check-ins, received 'strongly favorable' feedback. Mentored 4+ engineers across many projects, deployments, and onboardings. Created RFC templates, Stacked MR best practices etc. adopted by the team. Conducted technical interviews, led code reviews, and coordinated cross-team design discussions.

Match Group (OkCupid)

May 2020 – Sep 2021

Machine Learning Engineer

New York City

- Owned end-to-end ML pipeline for subscription discount optimization: feature engineering, model training, A/B testing, deployment, and production monitoring. Discovered high prediction variance across model runs; designed ensemble uncertainty estimation using 100-model bagging to quantify and stabilize outputs for production deployment.
- Achieved a 6% overall revenue increase through A/B tested pricing models.

FactSet

Apr 2015 – May 2020

ML Engineer → Senior ML Engineer

New York City

- Led multiple ML initiatives: (1) Speaker identification system for earnings calls using spectrograms and CNNs, (2) Private company fact extraction from 1.6M websites using ELMo/BiLSTM. Rewrote MLangID language identification service. Led machine translation infrastructure (Polish SMT achieving BLEU 69.10).
- Achieved a 20% reduction in human-hours for earnings call processing, automated extraction from millions of documents.
- Led team of 3 engineers on Document Screening: built autosuggestion and concept similarity systems. Created FingerPrinter deduplication service (10× response improvement: 1000ms→100ms). Architected Formula Lookup using distributed trie and n-gram language models on Spark.
- Improved formula ranking from 5.6 to 2.3, 66% faster document processing, powered StreetAccount trending news.
- † Technical Leadership
- Established engineering best practices: Jenkins CI, comprehensive test suites, documentation standards. Mentored new hires and
- junior engineers. FingerPrinter became the model Java project within the ML group.

Tata Research Development and Design Centre

Jul 2011 – Jul 2013

ML Research Engineer

India

- Wrote an algorithm based on Shape Context for finding frequently occurring patterns and events, achieving results comparable to SAX and DTW with 7% better performance in the domain of car sensors.
- Implemented an ETL framework that exploits the power of map-reduce and big databases to fuse incongruous enterprise data from disparate sources in near real-time.
- ⚙ Event Detection in Time Series (Java, Python, RapidMiner) [SVM - RBF] P

Projects

Listing IQ: Interactive CMA Platform / Django, DocumentDB, PyTorch

Link ↗ / GitHub ↗ / Kaggle ↗ / Google Scholar ↗

- Led proof-of-concept and core architecture for platform now live as Listing IQ CMA: AI-powered tool enabling agents to customize property comparisons with map-based filtering, editable valuations, and amenity-level explanations.

Zestimate Customer Care Agent + Reliability Infrastructure

/ React, LangChain, FastAPI

- Designed and shipped the first LLM assistant deployed to Zillow customer support: personalized Zestimate explanations grounded in real valuation data.

Active Listing Comps Engine / Apache Spark, Metaflow, H3 Geospatial Link ↗

- Architected and built similar listings comparison engine from ground up: batch pipeline and real-time API using H3 geospatial indexing and customizable ranking algorithms.

Next-Gen Zestimate: Multimodal + NAM + Explainability

/ PyTorch, CLIP, Databricks

- Leading Image Embeddings project: experimenting with DualLossAutoEncoder to predict property condition from CLIP embeddings for Zestimate integration.

Discount Optimization / Python, Keras, TensorFlow, Weights and Biases Link ↗

- Owned end-to-end ML pipeline for subscription discount optimization: feature engineering, model training, A/B testing, deployment, and production monitoring.

ML-Powered Financial Data Extraction / Python, TensorFlow, Keras, Sagemaker

- Led multiple ML initiatives: Speaker identification system for earnings calls using spectrograms and CNNs, Private company fact extraction from 1.6M websites using ELMo/BiLSTM.

Financial Document Search & Ranking Systems / Apache Spark, Java, Python

Google Drive ↗

- Led team of 3 engineers on Document Screening: built autosuggestion and concept similarity systems.

Event Detection in Time Series / Java, Python, RapidMiner Link ↗ / arXiv ↗ / Link ↗

- Wrote an algorithm based on Shape Context for finding frequently occurring patterns and events.

Technical Skills

Languages: Python, Java, Bash, SQL

AI/ML: RAG, Embeddings

Frameworks/Libraries: C/C++, LLM & GenAI, LangChain, LLM Guardrails, CLIP, Classical ML, PyTorch, TensorFlow, A/B Testing, Uncertainty Estimation, Model Monitoring, Data & Infra, PySpark, Metaflow, FastAPI, Django, Cloud & CI/CD, Leadership, System Design, Technical Interviews, Mentoring, RFC Authorship, Cross-team Coordination

Databases: Vector DBs, Pinecone, Databricks

Tools & Platforms: MLflow, Docker, Kubernetes, Terraform, AWS, GitLab CI, Weights & Biases

Leadership / Extracurricular

Organizer at MUFin Program committee member, paper reviewer at top ML conferences

Developer at Lotion

- Unofficial Notion.so Desktop app for Linux (2K+ GitHub stars / 60K+ Clones & Downloads)

Developer at Romadeva

- Tool to convert Roman script to Indic(Devanagari) script (Used by Translators Without Borders)

Developer at Quena

- Question and Answering system – Indexed 1.6 Million Wikipedia documents, designed a question parser and a ranking algorithm based on popularity. (Apache Solr, NER, POS tagger)

Publications

- Inferring Latent Attributes of an Indian Twitter user using Celebrities and Class Influencers, ACM Hypertext, 2015
- Inferring gender of a Twitter user using celebrities it follows, CORR, 2014
- Architecture for Automated Tagging and Clustering of Song Files According to Mood, IJCSI, 2010